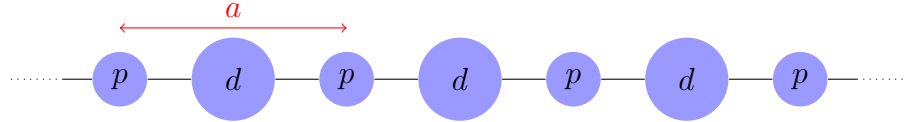


Homework 7

More tight-binding along with projected DOS

1. (50 pts) Here we will entertain a simple model, building on the last homework. Let us assume the 1d chain pictured below. We have one p orbital and one d orbital per unit cell, with on-site energies ϵ_p and ϵ_d , respectively. The only hopping parameter is the nearest neighbor hopping between p and d , denoted t_{pd} . Here we will have $\epsilon_d - \epsilon_p = 1$ and $t_{pd} = \frac{1}{4}$.



- Write an analytic equation for the eigenvalues as a function of the parameters and k , and make a band structure plot of the result in the first BZ.
 - Derive an analytic expression for the total DOS, and plot the result.
 - Compute an expression for the projected DOS of the p states and the d states. Plot these two curves, and please verify that they add up to the total DOS.
 - Given that $\epsilon_d - \epsilon_p > t_{pd}$, derive an analytic expression for the upper and lower band that is correct to order t_{pd}^2 . Please plot the result on top of the exact band structure from part a.
 - Assuming that there are 3 electrons per unit cell, compute the Fermi energy.
2. (50 pts) This problem will explore the low energy electrons of graphene. Consider a single p_z orbital on each site of the honeycomb structure of graphene. Assume that the on-site energy for this orbital is zero; and that there is only nearest neighbor hopping of $t = 2.5\text{eV}$.
- Derive an analytic expression for the Eigenvalues.
 - Plot the band structure along the edge of the IBZ $\Gamma \rightarrow M \rightarrow K \rightarrow \Gamma$.
 - Assuming 1 electron per site (ie. two per unit cell), find the Fermi energy.
 - Derive an analytic expression for dispersion around the K point.
 - Analytically compute the DOS near the Fermi energy using your analytic expression from the above. The expression for the DOS should depend on t .